
MODELING DEMOGRAPHIC TRENDS IN VIRGINIA USING MACHINE LEARNING

TECHNICAL REPORT

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ABSTRACT

This project investigates how machine learning can be used to model and predict demographic population trends in Virginia. So far, we have worked with population estimate data from Virginia Open Data Portal, which includes features such as year, age, sex, race, and ethnicity. After cleaning and pre-processing the data set, we applied multiple linear regression and random forest regression as baseline models to predict population estimates. These initial models have allowed us to establish early performance benchmarks. In future stages, we plan to explore additional regression techniques, conduct error analysis, and investigate which demographic features most influence population. As such, we plan to experiment with other models, look into feature importance, and run deeper error analysis.

1 Introduction

1.1 Motivation

Population trends prove to be important for understanding city planning, resource allocation, and political decision-making. This machine learning project aims to analyze Virginia's population growth (or decline) using regression, a supervised learning technique. We intend to predict future population changes and look at external factors that may be affecting their change. This can help civil engineers, politicians, and even economists make data-driven decisions for city development.

1.2 Related Works

Previous studies have applied population prediction with machine learning models. In a paper by Smith et al. (2020), his research group applied time series analysis and LSTMs to predict population growth in metropolitan areas. In a different study, Jones et al. (2021) used K-Means clustering to cluster the population based on socioeconomic factors, which shows how effective unsupervised learning can be in demographic analysis.

Our work builds on these approaches by applying supervised regression techniques to demographic population data at the locality and subgroup level within the state of Virginia. Unlike previous studies which focus on smaller geographic regions or local-level forecasting, our project explores broad demographic breakdowns (e.g., by age, race, and sex) to model and predict larger populations.

2 Methodology

2.1 Data Cleaning and Preprocessing

We worked with the Virginia Single-Race Population Estimates dataset from the Virginia Open Data Portal. The dataset includes annual population numbers grouped by multiple features including AGE, SEX, RACE, and LOCALITY. The target variable was ESTIMATE, the estimated population count.

Our preprocessing pipeline included the following.

- Removing redundant columns such as unnamed indices and codes.
- Filtering out rows with missing values in significant fields.
- Using One-hot Encoder to encode categorical variables like SEX and RACE.
- Scaling numerical features AGE and YEAR using `StandardScaler`.

We used `scikit-learn`'s `ColumnTransformer` and `Pipeline` to ensure consistent transformation across training and evaluation datasets.

2.2 Model 1: Linear Regression

At first, we implemented a Linear Regression model as a baseline. This model assumes a linear relationship between the input features and the population estimate. We performed 5-fold cross-validation to evaluate model performance across multiple train-validation splits. The results of this model served primarily as a benchmark for more complex models.

2.3 Model 2: Random Forest Regressor

To capture nonlinear patterns and feature interactions, we implemented a `RandomForestRegressor`. We performed 5-fold cross-validation and evaluated performance using Root Mean Squared Error (RMSE) and R-squared (R^2).

Due to resource constraints in Google Colab, we tuned the following hyperparameters for speed:

- `n_estimators` was set to 50 (number of trees)
- `max_depth` was capped at 10 (tree depth)

The Random Forest was configured to balance model complexity and training/testing speed, and was evaluated using the same cross-validation strategy.

2.4 Model 3: LSTM

Given that our dataset spans multiple years, we experimented with a Long Short-Term Memory (LSTM) model to better capture temporal relationships. Instead of using demographic features directly, we transformed the data into a time-series format:

- First, we pivoted the dataset into a matrix where each row represented a locality, and each column represented a year.
- The values in the matrix were the sum of population estimates across all demographic subgroups.
- Rows with missing values were dropped to ensure time continuity.

We then prepared the data as follows:

- `X` consisted of population data from 2010 to 2022.
- `y` was the target population for 2023.
- Both `X` and `y` were scaled using `MinMaxScaler`.
- `X` was reshaped to 3D format (samples, timesteps, features) required by Keras LSTM.

We trained a Keras LSTM model with the following architecture:

- LSTM layer with 64 units

- Dropout layer (rate 0.2)
- Dense layer with 32 ReLU-activated units
- Output Dense layer with 1 unit (population prediction)

The model was compiled with the Adam optimizer and Mean Squared Error (MSE) loss. Training was performed over 100 epochs with a batch size of 16. Validation loss was monitored using an 80/20 train/test split, and training history was visualized to determine if we would run into overfitting.

2.5 Evaluation Metric

All models were evaluated using Root Mean Squared Error (RMSE), with the formula as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE was an appropriate metric choice for population prediction due to the exponent penalizing large errors more heavily and its sensitivity to scale.

2.6 Tools and Environment

The entire analysis was performed in a Jupyter Notebook using Python 3. Libraries included but weren't limited to pandas, scikit-learn, tensorflow, seaborn, and matplotlib. All calculations were implemented on Google Colab.

3 Experiments

During implementation, we focused on performance, learning behavior, and error reduction. We designed separate experiments for the regression-based models (Linear Regression, Random Forest) and the time-series model (LSTM), using evaluation metrics such as RMSE and R^2 .

3.1 Baseline: Linear Regression

We began with a multiple linear regression model, primarily to serve as a performance benchmark. We used 5-fold cross-validation on the full dataset after preprocessing. For each fold, we recorded RMSE and examined residual patterns across subgroups. While simple and interpretable, it struggled to extract meaningful, non-linear relationships present in the data, which resulted in relatively high RMSE values.

Population peaks in certain age groups or nonlinear demographic shifts over time were just some of the complexities linear regression wasn't able to reconcile with. The predictions often underestimated population counts for large subgroups and overestimated for smaller ones, which indicated to us that a linear fit was insufficient and inappropriate as a solution.

3.2 Random Forest Regressor

To address the limitations of linear models, we implemented a Random Forest Regressor and repeated the 5-fold cross-validation procedure. We used a reduced number of estimators and a capped tree depth to ensure reasonable training time on Google Colab.

The Random Forest significantly outperformed the linear regression model in both mean RMSE and variance across folds, with an average RMSE score less than half compared to Linear Regression.

However, there was bias in prediction accuracy depending on population subgroup. Demographic clusters with more training examples, such as middle-aged and majority race groups, had a higher prediction accuracy, while underrepresented groups such as racial minorities generally corresponded with higher prediction errors. This showed that Random Forest was exploiting important patterns yet underfitting other parts of the distribution.

3.3 Feature Importance Analysis

In addition to evaluating overall model performance, we conducted a feature importance analysis using the trained Random Forest model. We first filtered features by correlation (for numeric) and mutual information (for categorical) to remove irrelevant inputs. This produced a reduced and interpretable feature set which was used to train the final model.

After, we calculated importance scores using the model's built-in `feature_importances_` attribute. The top four features were identified based on their contribution to prediction variance across trees.

3.4 LSTM

For the LSTM, the data was split into an 80/20 train/test split based on region. The model was trained for 100 epochs, and we tracked training and validation loss at each step. We visualized these trends to detect overfitting or instability.

3.5 Error Analysis Across Models

For each model, we conducted an error analysis to identify where and why predictions deviated most from the actual data. We found the following:

- **Extreme age groups** (e.g., under 5 and over 85) had noisier trends and higher variance in prediction error.
- **Small-population localities** were especially difficult to model due to inconsistent year-to-year changes and limited representation in training.
- **Demographic minorities** were often underrepresented in training data, leading to higher residuals in both the regression and forest-based models.

4 Results

All code can be found in this repository: <https://github.com/elliotschmango/Virginia-Population-Analysis>

4.1 Linear Regression

The multiple linear regression model was evaluated using 5-fold cross-validation. The RMSE values for each fold were as follows:

[595.96, 582.66, 598.27, 572.98, 585.48]

The average RMSE across all folds was 587.07, with a standard deviation of 9.22. This indicates the model's limited capacity to capture the nonlinear and interaction-based relationships inherent in the dataset.

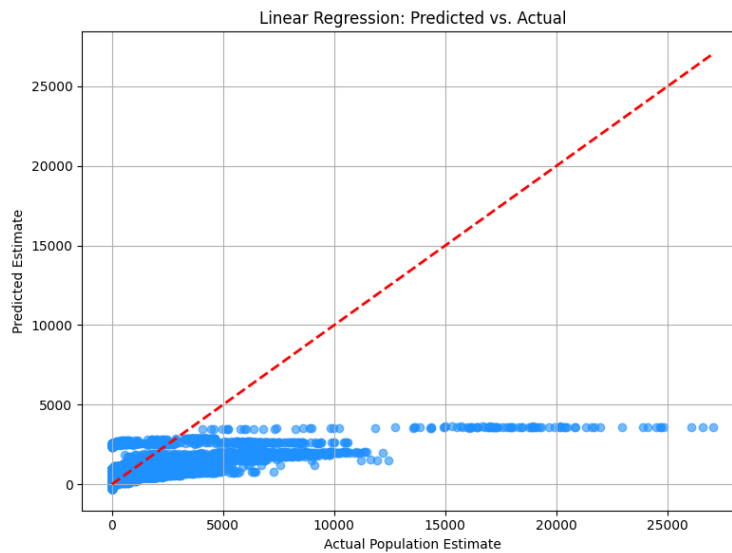


Figure 1: Comparison of Predicted vs Actual Population using Linear Regression

4.2 Random Forest

The Random Forest Regressor achieved significantly better performance:

- Root Mean Squared Error (RMSE): 266.31
- Coefficient of Determination (R^2): 0.8663

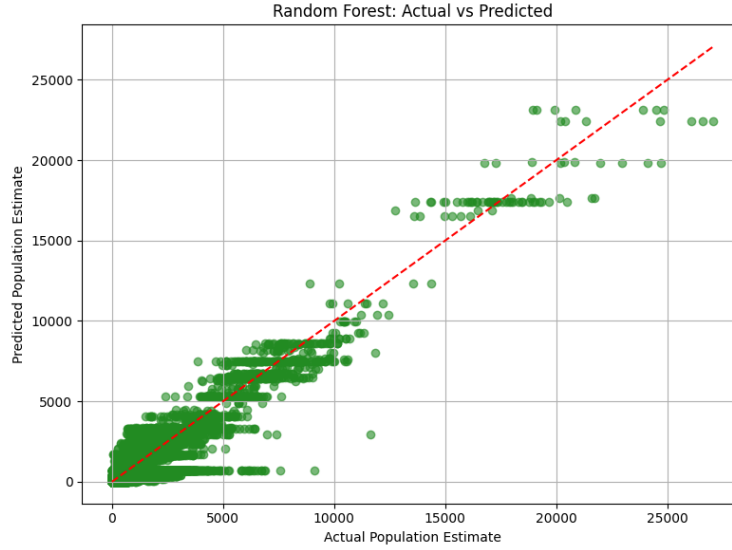


Figure 2: Comparison of Predicted vs Actual Population using Random Forest Ensemble

4.3 Feature Importance (Random Forest)

To interpret which inputs most influenced the model's predictions, we extracted feature importance scores from the trained Random Forest. The top four most important features were:

- LOCALITY_Fairfax County
- TIERED_RACE_ETHNICITY_White
- DISTRICT_NAME_Virginia Beach
- LOCALITY_Chesterfield County

These features included a combination of geographic identifiers and race-based subgroup encodings, suggesting that population prediction is strongly influenced by location and subgroup-specific dynamics.

The importance scores also reflected the relative predictive power of numerical versus categorical inputs:

- AGE and YEAR were consistently retained during filtering, indicating strong correlation with the target variable.
- One-hot encoded race and sex groups provided substantial predictive signal, particularly for majority demographics.
- Categorical variables with low mutual information were excluded, which helped reduce model noise and improve stability.

4.4 LSTM

The LSTM model was tested on a held-out set of localities. Performance metrics were:

- Root Mean Squared Error (RMSE): 14,049.22
- Mean Absolute Error (MAE): 12,976.76

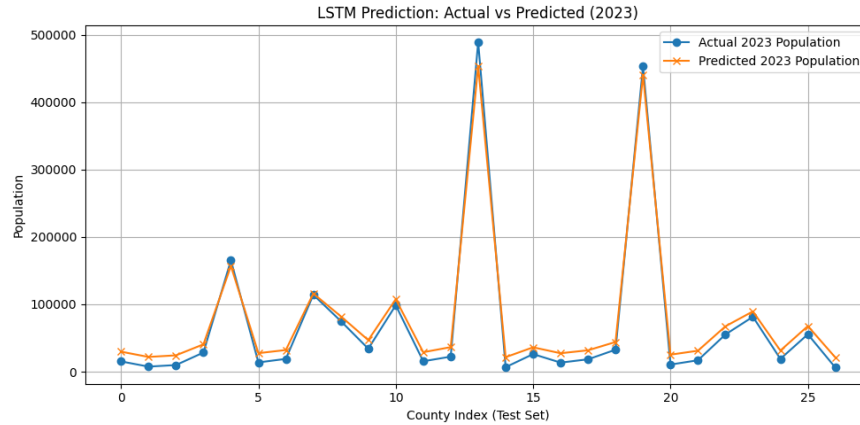


Figure 3: Comparison of Predicted vs Actual Population using LSTM

Although the LSTM was designed to capture time-dependent relationships, its error was an order of magnitude higher than that of the Random Forest. This performance gap may be due to limited sequence length, aggregation of input features, or challenges in modeling sharp population changes with a small dataset.

5 Conclusion

Our aim in this project was to see the extent to which machine learning could be used to predict population estimates across Virginia using annual demographic data. Our hypothesis was that advanced models (particularly ensemble and neural networks) can improve predictive accuracy over simpler baselines and provide meaningful insights into population trends that could aid in state-level planning. Our results confirmed this hypothesis. The Random Forest model, which captured complex interactions between variables such as age, race, and locality, significantly outperformed linear regression, achieving an RMSE less than half that of the baseline model. Although the LSTM model didn't perform as well as expected, it was able to show the merit of time-series forecasting with chronological data.

These findings have real-world implications for Virginia's communities and governance. Accurate population estimates support better resource allocation in public health, education, and infrastructure and help shape policy. For instance, predict shrinking populations can encourage economic development initiatives to mitigate population loss. In addition, identifying demographic groups that are harder to model highlights the need that underserved communities should not overlooked in planning decisions.

That is not to say that our analysis also surfaced limitations. The models struggled to make accurate predictions for localities with small or highly variable populations, and for underrepresented racial groups. This points to data sparsity and bias—challenges that must be addressed to build fairer and more inclusive forecasting systems. Future work could extend this analysis by incorporating sociological variables such as economic indicators, housing development, or migration trends to improve forecasting.

6 Contributions

Elliot:

1. Wrote project checkpoint
2. Data exploration/cleaning
3. Implemented Linear Regression
4. Implemented LSTM
5. Video editing
6. Planned/Wrote final report

Thomas:

1. Wrote project proposal

2. Implemented Random Forest Ensemble
3. Drew conclusions from results
4. Wrote video script
5. Planned final report

References

- [1] Jones, M., Davis, T., Chen, A. (2021). *Demographic Clustering of Regional Populations Using K-Means*. International Conference on Data Science, 8(1), 22–34.
- [2] Smith, J., Zhang, L., Patel, R. (2020). *Forecasting Urban Population Growth Using Recurrent Neural Networks*. Journal of Urban Analytics, 12(3), 45–60.
- [3] Virginia Department of Health. (n.d.). Virginia single-race population estimates. Virginia Open Data Portal. <https://data.virginia.gov/dataset/vdh-virginia-single-race-population-estimates>